

Lily Feldman

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EDUCATION

University of Michigan - College of Engineering

MS in Robotics

BSE in Robotics, Dean's List, University Honors, GPA: 3.6 / 4.0

Ann Arbor, MI

Aug. 2026-May 2027

Aug. 2023-May 2026

Selected Courses: Robot Control, Electronic Circuits, Sensors & Signals, Robotic Prototyping and Kinematics, Programming and Intro to Data Structures, Self-Driving Cars, Computational Linear Algebra, Differential Equations, Dynamics & Vibrations, Probabilistic Methods in Engineering, Human-Robot Interactions, Electricity & Magnetism, Multivariable Calculus, Discrete Math

Technical Skills: *Computational and Programming*– Python, C++, C, MATLAB, Java, Julia, Git, Shell, HTML; *Design*– AutoCAD, SolidWorks, Onshape; *Hardware*– circuitry design, microcontroller implementation (Arduino, Raspberry PI), iterative prototyping

WORK EXPERIENCE

Computational Medicine Laboratory, New York University

May. 2025 – Aug. 2025

Researcher, Tandon Undergraduate Summer Research Program

New York, NY

- Designed and built an autonomous closed-loop wearable medical device with a flexible actuator band geared towards Long Covid symptom management and mental health research
- Engineered and implemented a cascade control system using both PID and fuzzy logic controllers to adaptively modulate actuator response in real time
- Integrated multimodal sensors, including skin conductance sensors and IMUs, and a piezoelectric pump system for robust bio signal acquisition
- Prototyped and tested the full system, honing expertise in CAD design and embedded microcontroller programming

KSCJ Consulting, Civil and Environmental Engineering Consulting

Mar. 2023 – Aug. 2023

Civil Engineering Intern

Armonk, NY

- Assisted engineers in performing comprehensive site inspections, evaluating environmental conditions, and assessing compliance with regulatory standards for land development projects
- Enhanced technical proficiency in CAD software (SolidWorks) by analyzing field plans to devise recommendations on vegetation removal and replacement, aiming to optimize project feasibility and mitigate negative environmental impacts
- Enriched valuable interpersonal and professional skills by engaging with engineers, office managers, and inspectors, facilitating effective teamwork and communication

PROJECT EXPERIENCE

Michigan Carbon Capture

Aug. 2024 – Present

Mechatronics Engineering Sub Team Leader

Ann Arbor, MI

- Aim to design a wastewater based microbial electrolysis cell to generate hydrogen and calcium carbonate as a means to remove carbon dioxide from the atmosphere
- Lead a team to engineer and implement electrical circuits for system operation, leveraging Arduino microcontrollers for precise sensor integration and data acquisition
- Develop and optimize an anode and cathode system to facilitate key chemical reactions, supporting the production of hydrogen and calcium carbonate
- Collaborate with various sub-teams to integrate biological, chemical, mechanical, and electrical engineering principles for the development of a cohesive system

Microrobot Swarm Control (Capstone Project)

Aug. 2025 – Dec. 2025

Mechatronic Design Engineer

Ann Arbor, MI

- Designed and built a magnetic actuation system to control microrobot swarms over a 1 ft × 1 ft workspace, improving scalability compared to traditional electromagnetic arrays
- Integrated full-stack robotics system (mechanical, electrical, software) and validated performance through experimental testing
- Engineered a custom magnetic end-effector and validated system performance through experimental testing of localization accuracy and control error
- Developed a camera-based perception pipeline using OpenCV to detect and localize microrobot swarms in real time
- Implemented closed-loop control and path planning algorithms to autonomously navigate microrobots through obstacle-filled environments

ADDITIONAL ACTIVITIES

Society of Women Engineers, Alpha Epsilon Phi Sorority, Michigan Songwriters Collective, Recreational Soccer